Practice workbook · How long is a vector? Magnitudes and finding unknowns

Working out the length of a vector with Pythagoras on its components (or the cosine rule for non-perpendicular base vectors), and reversing this to find an unknown constant when a length is given.

7 question parts: 1 worked-example retry and 6 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-VEC-02

1. Question 4 · part b · 3 marks

2020 JAN · Paper 02

Worked-example retry · Attempt part b only.

4 The points A, B, C and D are the vertices of a quadrilateral $ABCD$ such that

$\overrightarrow{AB} = 7\mathbf{i} + p\mathbf{j}$ $\overrightarrow{AC} = 11\mathbf{i} - p\mathbf{j}$ $\overrightarrow{AD} = 4\mathbf{i} - 2p\mathbf{j}$

(a) Show that, for all values of p, $ABCD$ is a parallelogram.

(3)

Given that $|\overrightarrow{BD}| = 3\sqrt{10}$

(b) find the possible values of p.

(3)

Given that $p > 0$

(c) find a unit vector which is parallel to $\overrightarrow{BD}$.

(1)

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1. Question 4 · part b · 3 marks · Working and answer

Working and answer

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2. Question 1 · part b · 1 mark

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

1 The position vector of the point A is $(3\mathbf{i} - 2\mathbf{j})$, referred to a fixed origin O.

The point B is such that $\overrightarrow{AB} = (6\mathbf{i} + 8\mathbf{j})$

(a) Find the position vector of B as a simplified expression in terms of $\mathbf{i}$ and $\mathbf{j}$

(2)

(b) Find the magnitude of vector $\overrightarrow{AB}$

(1)

(c) Find a unit vector, in terms of $\mathbf{i}$ and $\mathbf{j}$, that is parallel to $\overrightarrow{AB}$

(2)

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2. Question 1 · part b · 1 mark · Working and answer

Working and answer

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3. Question 4 · part a · 4 marks

2023 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

4 O, A and B are fixed points such that

$$\overrightarrow{OA} = 5\mathbf{i} + 7\mathbf{j} \qquad \overrightarrow{AB} = a\mathbf{i} + 16\mathbf{j} \qquad \text{and} \qquad \left|\overrightarrow{OB}\right| = 5\sqrt{29}$$

(a) Find the possible values of a

(4)

Given that $a > 0$

(b) find a unit vector that is parallel to $\overrightarrow{AB}$

(2)

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3. Question 4 · part a · 4 marks · Working and answer

Working and answer

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4. Question 3 · part a · 4 marks

2022 JUN · Paper 02

Further practice · Attempt part a only.

3 O, A and B are fixed points such that

$$\overrightarrow{OA} = (p\mathbf{i} - 4\mathbf{j}) \qquad \overrightarrow{OB} = \mathbf{i} + (2p + 1)\mathbf{j}$$

Given that $\sqrt{2}\left|\overrightarrow{OA}\right| = \left|\overrightarrow{OB}\right|$ and $p > 0$

(a) find the value of p

(4)

Using this value of p

(b) find a unit vector that is parallel to $\overrightarrow{AB}$

(5)

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4. Question 3 · part a · 4 marks · Working and answer

Working and answer

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5. Question 3 · part a · 4 marks

2024 JUN · Paper 02 (regional R)

Further practice · Attempt part a only.

3 O, A and B are fixed points such that

$$\left|\overrightarrow{OA}\right| = 3\sqrt{5} \qquad \overrightarrow{AB} = \mathbf{i} + 3a\mathbf{j} \qquad \overrightarrow{OB} = 7\mathbf{i} + 2a\mathbf{j}$$

Given that $a > 0$

(a) find the value of a

(4)

(b) Hence find a unit vector parallel to $\overrightarrow{OA}$

(2)

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5. Question 3 · part a · 4 marks · Working and answer

Working and answer

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6. Question 6 · part a · 4 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part a only.

6 Referred to a fixed origin O, the position vector of the point A is $(9\mathbf{a}+12\mathbf{b})$ and the position vector of the point B is $(2\mathbf{a}+p\mathbf{b})$ where $|\mathbf{a}|=|\mathbf{b}|=1$

Given that $|AB|=25$

(a) show that the two possible values of p are -12 and 36

(4)

For $p=-12$

(b) find a unit vector parallel to $\overrightarrow{OB}$
Give your answer in terms of **a** and **b** in its simplest form.

(2)

Point C lies on OB such that $OC:CB=2:3$

For $p=36$

(c) find $\overrightarrow{CA}$
Give your answer in terms of **a** and **b** in its simplest form.

(3)

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6. Question 6 · part a · 4 marks · Working and answer

Working and answer

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7. Question 8 · part a.ii · 3 marks

2024 NOV · Paper 02

Further practice · Attempt part a.ii only.

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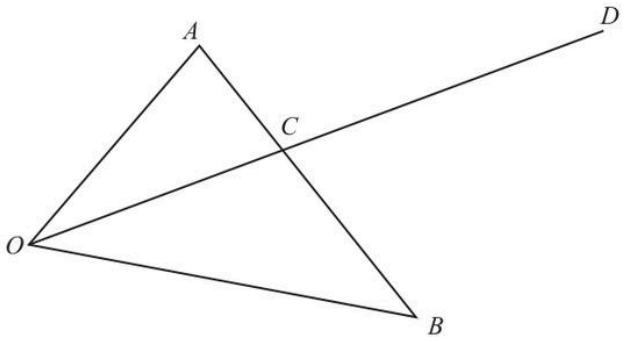


Diagram **NOT** accurately drawn

Figure 2

Figure 2 shows triangle AOB

$\overrightarrow{OA} = 4\mathbf{a} + 5\mathbf{b}$ $\overrightarrow{OB} = 8\mathbf{a} - \mathbf{b}$ $\overrightarrow{OD} = 15\mathbf{a} + 10\mathbf{b}$ where $|\mathbf{a}| = |\mathbf{b}| = 1$

(a) (i) Find $\overrightarrow{AB}$ in terms of **a** and **b**

(ii) Find, in its simplest form, the exact value of $|\overrightarrow{AB}|$ **(3)**

(b) Find the area of triangle AOB **(4)**

The point C lies on AB and OD such that O, C and D are collinear.

(c) Use a vector method to find vector $\overrightarrow{OC}$ as a simplified expression in terms of **a** and **b** **(5)**

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7. Question 8 · part a.ii · 3 marks · Working and answer

Working and answer

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